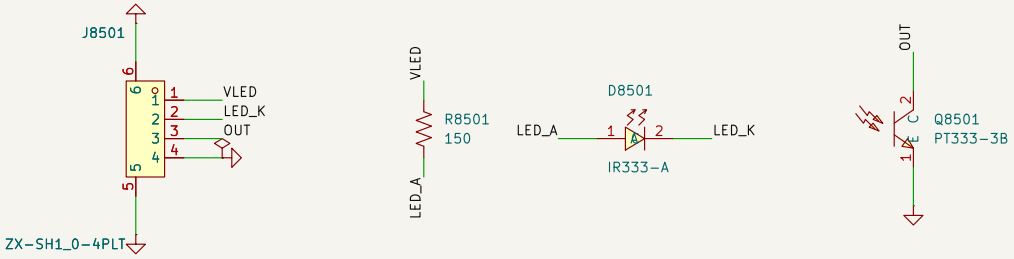
OOMWOO cliff sensor board (rev 1, 2026-10-28). One design, fitted 6x. No lenses: narrow-beam 5 mm parts (IR333-A 20 deg,
PT333-3B black lens) tilted toward each other so their beams cross at the floor. The tilt sets the reference distance:
crossing depth = (spacing / 2) / tan(tilt), e.g. 18 mm spacing: 30 deg -> 15.6 mm, 25 deg -> 19.3 mm, 20 deg -> 24.7 mm.
Beyond the crossing point the beams diverge, so a far floor (a cliff) returns little light even if it is white.
The main board switches the LED cathode (CLIFF_LED_EN, modulate it to cancel ambient light) and pulls OUT up (47k).
Refs 85xx: unique on the OOMWOO panel.



J8501 to the main board, pin 1 to pin 1:
1 VLED (5 V), 2 LED cathode (main board
switches it), 3 OUT, 4 GND

Assembly (JLC): D8501 and Q8501 THT with the assembly standoff (~2 mm) so the leads can be bent;
pin bending to the tilt angle, or bend by hand in the 3D-printed housing. LED: flat side = cathode.
Q8501: flat side / square pad = emitter. Spacing D8501-Q8501: 18 mm centre to centre.

OOMWOO cliff sensor

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Title: Cliff sensor board (fit 6x)

Size: USLetter	Date: 2026-10-28	**Rev: 1**
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